

Stalin goes to Vegas

Problem ID: stalininvegas

Stalin has heard much of life in the states, and has decided to pay them a visit. In order to experience the true glory of the American lifestyle he has decided a trip to Vegas is in order. While Stalin disagrees with how wealth gets distributed by the casinos he is fascinated by the cards. However, unfortunately for the cards, Stalin really prefers when everyo... everything stays in order. He asks his most loyal servant(you) to sort the cards, "disposing" of any that dare step out of line.

The rules are as follows: The cards are layed out in a random order. You have an ordered list of cards in your hand, starting out empty. On each turn you must either take or discard the left-most or right-most card. You can only take a card if its face value is less than your lowest value card, or greater than your greatest value card. The goal is to have as many cards in your hand as possible in the end.

Stalin awaits, do not disappoint.

Input

The input consist of one test case. The first line contains 1 integer n , such that $1 \leq n \leq 52$ (A full deck of cards). The second line consists of n space-seperated integers, from 1 to n , the cards that are to be sorted.

Output

Output the maximum number of cards that can be successfully sorted, meaning the number of cards that do not get discarded.

Sample Input 1

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5
5 3 2 4 1
```

Sample Output 1

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4
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Sample Input 2

```
6
1 2 3 4 6 5
```

Sample Output 2

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6
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